Hyperspectral and LiDAR Data Classification Using Joint CNNs and Morphological Feature Learning

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***Abstract*— Convolutional neural networks (CNNs) have been extensively utilized for hyperspectral image (HSI) and light detection and ranging (LiDAR) data classification. However, CNNs have not been much explored for joint HSI and LiDAR image classification. Therefore, this article proposes a joint feature learning (HSI and LiDAR) and fusion mechanism using CNN and spatial morphological blocks, which generates highly accurate land-cover maps. The CNN model comprises three Conv3D layers and is directly applied to the HSIs for extracting discriminative spectral–spatial feature representation. On the contrary, the spatial morphological block is able to capture the information relevant to the height or shape of the different land-cover regions from LiDAR data. The LiDAR features are extracted using morphological dilation and erosion layers that increase the robustness of the proposed model by considering elevation information as an additional feature. Finally, both the obtained features from CNNs and spatial morphological blocks are combined using an additive operation prior to the classification. Extensive experiments are shown with widely used HSIs and LiDAR datasets, i.e., University of Houston (UH), Trento, and MUUFL Gulfport scene. The reported results show that the proposed model significantly outperforms traditional methods and other state-of-the-art deep learning models. The source code for the proposed model will be made available publicly at https://github.com/AnkurDeria/HSI+LiDAR.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI) classification, light detection and ranging (LiDAR).**

I. Introduction

HYPERSPECTRAL images (HSIs) convey rich spectral information, which is encoded in hundreds of narrow and contiguous spectral bands [1], [2]. The robust discriminative behavior of the spectral features helps to capture subtle spectral differences, and thus, the classification task with HSIs has become one of the key aspects in several applications related to Earth observation, such as deforestation, land-cover mapping, and mineral exploration (to mention a few) [3]–[5]. However, HSIs are unable to properly differentiate between two complex land covers as long as they produce similar spectral responses. To illustrate this point, we plot some spectral signatures from the well-known University of Houston HSI dataset. It can be observed that classes such as Healthy Grass have spectral signatures that are similar to those of Stressed Grass, as shown in Fig. 1. Other classes such as Highway, Parking Lot 1, Railway, Residential, and Road also exhibit very similar spectral signatures. This poses a challenge in the classification of HSIs, as the classifiers struggle to distinguish similar land covers. There often exist such types of complex land-cover types in urban and rural areas, and this problem can be easily overcome by using another source of information, obtained from light detection and ranging (LiDAR) data.

LiDAR data can provide elevation information of height and shape of the image surface with respect to the sensor [6], thus providing complementary information for HSI data, and can, therefore, be useful to distinguish the land covers that are made up of the same material. Using LiDAR, one can accurately classify between roofs and roads even when they have similar materials. However, LiDAR data fail to distinguish between the classes in which the objects contain similar elevation information, for example, two roofs having similar elevation information with different materials, i.e., concrete or asphalt. Developing HSI and LiDAR data fusion techniques represents an interesting and challenging research topic to explore, whose performance has already been studied in the literature for land-cover classification [6]–[8].

In the early days, traditional methods were widely used for HSI classification, which showed significant performance gains even with limited training [9]–[11]. These methods extract and learn the feature representation and then fit those features into a machine learning model for classification [1], [8], [12]–[14]. Among the traditional models, the support vector machine (SVM) with a nonlinear kernel is widely used one, especially when the training samples are limited [15], [16]. To further improve the performance, some handcrafted features are designed before the use of the classifier for unlabeled HSI data [17]. However, the performance achieved was not satisfactory due to the nonlinear relation between the captured spectral information and that of the corresponding

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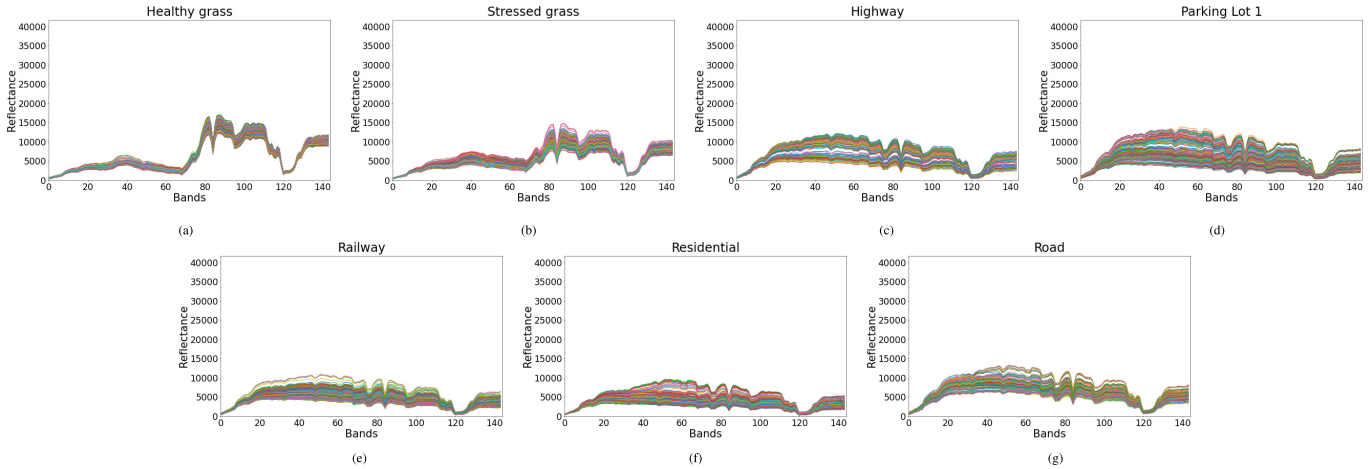


Fig. 1. Classwise spectral signatures of the classes such as Healthy grass, Stressed grass, Highway, Residential, and Road extracted from the University of Houston (UH) hyperspectral dataset. (a) Healthy grass. (b) Stressed grass. (c) Highway. (d) Parking Lot 1. (e) Railway. (f) Residential. (g) Road.

object of HSI data, which makes accurate classification more challenging for traditional methods. Zhang *et al.* introduced polygonal partition to capture geometrical shape features of land covers in real-life environments and adopted spectral self-similarity criteria for enlarging the number of training samples, which led to better classification performance in limited training scenarios [18]. Duan *et al.* [19] extracted discriminative feature structural profiles from the raw HSIs and utilized a decision label fusion strategy for accurate classification. Kang *et al.* [20] introduced redundancy-free and discriminative HSI features based on the intrinsic image decomposition strategy for improved classification.

Recently, deep learning models have shown great success in land-use and land-cover (LULC) classification tasks and outperformed traditional methods due to their automatic capability to achieve high- and low-level feature learning using its trainable kernels. For instance, convolutional neural networks (CNNs) extract feature maps that are invariant to the local changes with respect to their input [21]. A CNN framework that extracts the contextual feature representation to capture more discriminative spectral information and spatial information was introduced by Lee and Kwon [22]. Hong *et al.* considered the spatial relationship between pixels by measuring spectral similarities and proposed a miniGCN for HSI classification, achieving the state-of-the-art classification accuracies [23], [24].

Due to their larger depth, most of the deep networks suffer from the vanishing gradient problem, and to tackle this, a residual network (ResNet) is introduced, which can ensure minimum information loss after each convolutional operation [25]. Zhong *et al.* [26] proposed a 3-D spectral–spatial residual network (SSRN) for better modeling spectral and spatial features. Extended models are proposed based on lightweight spectral–spatial squeeze-and-excitation attention [27], adaptive kernels [28] and pyramidal residual networks [29], lightweight heterogeneous kernel convolution [30], and gradient centralized convolution [31], [32], which enables efficient feature extraction and classification. To improve the former models, Roy *et al.* proposed a sequential model to establish a relation between the extracted feature from 3-D and 2-D convolutions [33]. Recently, Ahmad *et al.* explored the attention mechanism with a hybrid dense network with different kernels in each block to enhance the feature learning for HSI classification [34]. In order to capture the shape information of arbitrary land-cover texture classes, morphological dilation and erosion layers with trainable kernels are employed with its trainable kernels for accurate classification of HSIs [35].

Though the HSIs exhibit rich spectral–spatial information, the lack of elevation information led to poor classification performance in a complex scene. To overcome this issue, LiDAR data can be taken into the consideration, which helps to better characterize the elevation information for the same survey area [36]. LiDAR data have also been studied for feature detection and extraction tasks [7], [37]–[39]. However, in order to utilize the complementary information between HSI and LiDAR data, a number of works have been proposed [40].

The joint use of HSI and LiDAR data can significantly improve the classification performance by effectively addressing the shortcomings of each of them and compared to the individual source of information [41]. In the beginning, most of the researchers applied morphological extinction profiles (EPs) and attribute profiles (APs) for joint feature extraction for accurate classification of HSIs and LiDAR data [42]–[44]. Similarly, Rasti *et al.* [45] improved the joint extraction of EPs by applying the total variation component analysis for feature fusion. Merentitis *et al.* [46] introduced the automatic fusion technique using the random forest classifier for joint HSI and LiDAR data classification.

Nevertheless, nowadays, deep learning is being extensively used for HSIs and LiDAR feature fusion [47] and classification. An earlier attempt was made by considering LiDAR as additional spectral bands of HSIs and fed together into a CNN model to learn both the features and perform classification tasks [48]. Ghamisi *et al.* [6] proposed a CNN model that extracts traditional features, i.e., EPs jointly from HSIs and LiDAR data that are fused and then fed to the CNN for classification. Chen *et al.* [49] introduced an end-to-end feature fusion CNN model that can perform three different tasks together, i.e., two separate CNNs are employed for feature extraction directly from the HSIs and LiDAR data, and then, extracted features were fused using concatenation and fed into a fully connected (FC) layer for final classification.

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Likewise, Wu *et al.* [50] explored the spectral–spatial CNN model for extracting discriminative HSI features and a spatial CNN model to extract the information from another modality data. Finally, both sources of information are concatenated, and then, the fused features are classified using FC layers. To reduce the network parameters, Hang *et al.* [8] proposed an efficient HSI and LiDAR data fusion technique using a shared CNN model and achieved performance gains compared to the existing. To solve the spatially fragmented classification problem, Zhao *et al.* [12] utilized hierarchical random walk in the CNN layer to learn the spatial consistency between HSI and LiDAR data for accurate classification. Hong *et al.* [51] designed a novel semisupervised deep cross-modal network by fully considering the unlabeled samples with the application to the classification of multimodal remote sensing data. Fang *et al.* [52] proposed an effective spatial–spectral cross-modality enhancement module for joint HSI and LiDAR data classification. Recently, Roy *et al.* [53] first introduced the multimodal fusion transformer (MFT) model to learn the abstract feature representation from both HSI and LiDAR data by utilizing the classification (`CLS`) token for land-used and land-cover classifications.

Morphology has been successfully applied in remote sensing long back to extract spatial information based on the concept of MPs and APs [54]–[57]. MPs and APs consist of a number of handcrafted features (e.g., height, area, volume, diagonal of the bounding box, and standard deviation) extracted by sequentially applying dilation and erosion operations with a set of structuring elements (SEs). Nevertheless, MPs and APs do have a few common limitations. First, the shape of the SE is fixed. Second, the SEs are only able to extract information related to the size of existing objects and fail to capture shape information of the arbitrary object boundary for complex regions, especially for LiDAR data. To overcome these limitations, a spatial morphological block is introduced, which uses its own trainable SEs in dilation and erosion layers to extract the robust and discriminative elevation information from LiDAR data. At the same time, a 3-D CNN model is employed for extracting the spectral–spatial features from HSIs. Finally, the two sources of features are combined using concatenation operation to train both models jointly in an end-to-end fashion. To summarize, the following key contributions have been made in this article.

1) In order to take advantage of both the morphological dilation and erosion operations, a novel spatial morphological convolutional block with trainable kernels is proposed to extract robust and discriminative elevation information from LiDAR data.
2) The spectral–spatial features can be learned from HSI data using a three-layered CNN that comprises Conv3D layers, whereas the elevation information can be better approximated from LiDAR data using trainable spatial morphological blocks, and both features are concatenated using standard fusion, i.e., channel concatenation, during training.
3) The spectral–spatial features from HSI data and the elevation information from LiDAR data can efficiently be learned using a joint training of a CNN and a spatial morphological convolutional network for accurate classification of land covers.
4) Extensive experiments are conducted using two disjoint datasets, i.e., University of Houston and Trento, where the training-test samples are mutually exclusive, 5% random training-test samples from the MUUFL dataset are selected, and the proposed method performs best in all three of them.

The rest of the article is organized as follows. The preprocessing of HSIs and LiDAR data is described in Section II. Section III introduces the proposed joint CNNs and the morphological feature learning network. The experimental results with detailed discussion are given in Section IV. Finally, Section V concludes the article with some remarks and hints at plausible future research lines.

II. Preprocessing of HSI and LiDAR Data

Let us consider a spectral–spatial HSI $\mathbf{X_H} \in \mathcal{R}^{M\times N\times B}$ and the corresponding LiDAR image $\mathbf{X_L} \in \mathcal{R}^{M\times N}$ where both represent the same spatial region over the Earth's surface. Here, the width M and the height N, respectively, represent two spatial dimensions for both images, and B refers to the spectral bands of the HSI image.

All the pixels are classified into C land-cover classes denoted by $C = (y_1, y_2, \ldots, y_c)$. The pixel $\mathbf{x}_{i,j} \in \mathbf{X_H}$, where $i = 1, \ldots, M$ and $j = 1, \ldots, N$, and hence, we can define the land-cover pixels as a spectral vector $\mathbf{x}_{i,j} = [x_{i,j,1}, \ldots, x_{i,j,B}] \in \mathcal{R}^B$ containing B number of elements. Similarly, to incorporate the spatial information, patch extraction is performed in a preprocessing step, where a spectral–spatial cube $\mathbf{x}_{i,j} \in \mathcal{R}^{k\times k\times B} \in \mathbf{X_H}$ from the normalized HSI data $\mathbf{X_H}$ and spatial cube $\mathbf{x}_{i,j} \in \mathcal{R}^{k\times k} \in \mathbf{X_L}$ from the LiDAR image with the neighboring regions of size $(k \times k)$ are extracted, centered at the target pixel (i, j).

Jointly exploiting spatial–spectral information can increase the discriminative power of feature learning network, and for that reason, spectral–spatial cubes $\mathbf{x}_h^{(i,j)}$ are extracted from raw HSIs and stacked into X_H prior to the feature extraction process. Similarly, for the LiDAR data, the spatial patch of the same size $\mathbf{x}_l^{(i,j)}$ is extracted and stacked into X_L before the actual feature extraction process. Finally, the training and test samples in each class for X_H and X_L can be represented by

$$\begin{aligned} D^{\text{train}} &= \left\{(x_h, x_l), y^{(i)} | i = 1, \ldots, P\right\} \\ D^{\text{test}} &= \left\{(x_h, x_l), y^{(i)} | i = 1, \ldots, Q\right\} \end{aligned} \tag{1}$$

where $x_h \in X_H$ and $x_l \in X_L$ are the samples randomly chosen from HSIs and LiDAR data. P and Q represent the number of training and test samples, and $y^{(i)}$ is the actual class for the ith targeted land-cover pixels.

III. Proposed Methodology

For better understanding, the core idea proposed (i.e., HSI + LiDAR classification) in this article has been shown in Fig. 2. The proposed HSI + LiDAR classification framework consists of a three-layered 3-D CNN (`Conv3D`) for discriminative HSI feature extraction and, similarly, a dual-branch morphological 2-D network to extract the robust spatial information from

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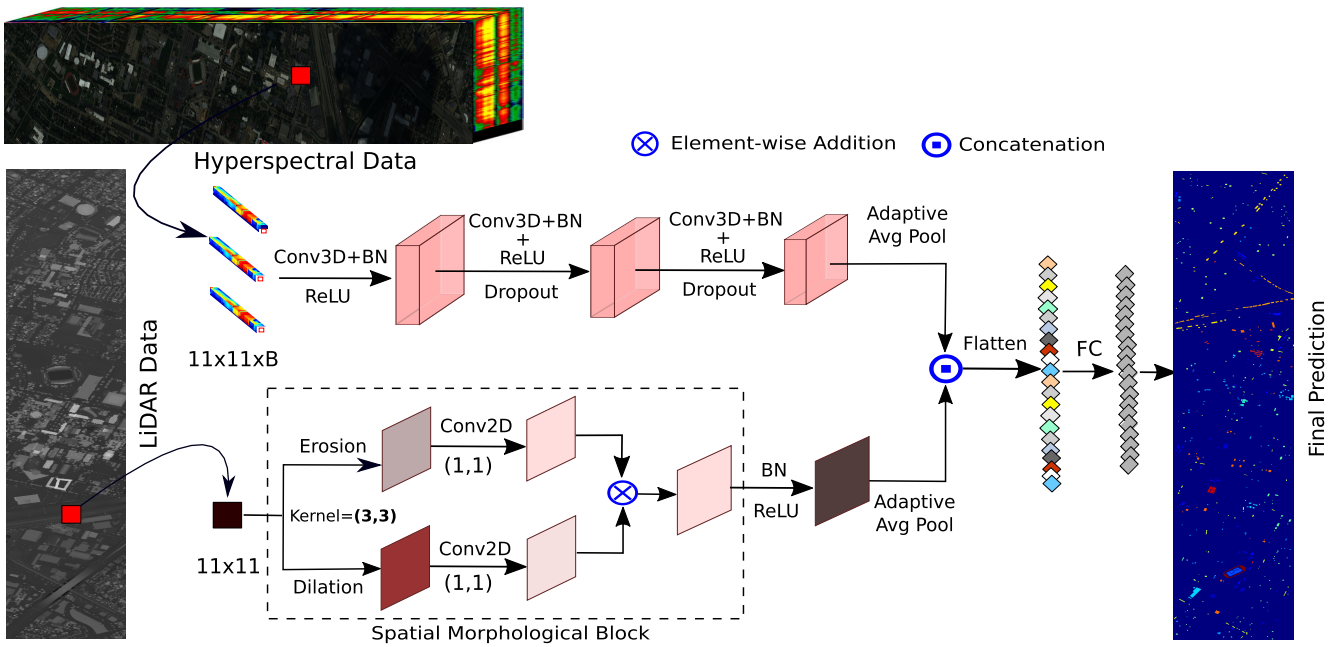


Fig. 2. Graphical representation of the proposed hyperspectral and LiDAR data fusion network.

the LiDAR data. Both the extracted features are fused using an addition ($\otimes$) operation immediately after the adaptive average pooling layer, which is also followed by a classifier that includes an FC layer with cross-entropy loss.

For the HSI, initially, channelwise normalization is performed without affecting the spectral dimension of the HSI data, and then, small cube patches of sizes $11 \times 11 \times B$ are extracted surrounding the given target pixel. Similarly, the image patches of the same spatial position as the HSI data are directly extracted from LiDAR data. The goal is to effectively combine both the information from $\mathbf{X_H}$ and $\mathbf{X_L}$ using a feature fusion step to improve the classification performance.

A. Hyperspectral Feature Learning via 3-D CNNs

CNNs exhibit promising performance in large-scale image classification tasks. The availability of enormous volumes of HSIs enables us to utilize the benefits of CNNs. The HSI $\mathbf{X_H}$ often exhibits a nonlinear relationship between the spatial information captured by the HSI sensors and the corresponding object of HSI data. This type of nonlinear characteristic will be more visible when dealing with $\mathbf{X_L}$.

The spectral–spatial behavior of HSIs always demands spectral and spatial feature learning for discriminative and robust classification. It has already been proved that CNNs are capable to extract high-level abstract features, which are also invariant to the source of data modalities, i.e., HSI and LiDAR datasets.

To extract robust and discriminative spectral and spatial features from raw HSIs, three layers of 3-D CNN are used to model the network. Fig. 2 shows the overall graphical representation of the Conv3D network for HSIs. The HSI cube $\mathbf{X}_H$ of size $(11 \times 11 \times B)$ is passed through the three consecutive Conv3D layers, and the sizes of the considered kernels of the first two layers are chosen to be $(3 \times 3 \times 7)$, whereas the size of the used kernel in the last Conv3D layer is $(3 \times 3 \times 32)$.

The shapes of the output feature maps after the first and second Conv3D layers are $(11 \times 11 \times 69)$ and $(11 \times 11 \times 32)$, respectively. Similarly, the output of shape $(11 \times 11 \times 1)$ is produced by the last Conv3D layer and to keep the shape 1 in the third dimension of the output feature maps after the last Conv3D layer; the third dimension of the kernel is chosen to be equal to the number feature maps, which is 32.

The convolutional layers utilize the feature padding mechanism to ensure the same shape and spatial size for both input and output feature maps. The batch normalization (BN) layer is used after every Conv3D layer to regularize the problem of overfitting due to the shortest of training samples and accelerate the training performance. ReLU activation is used to introduce the nonlinearities into the learning process during the training of the network. The last two Conv3D layers employed dropout in addition to the BN, which greatly reduces the effects of the vanishing gradient problem.

B. LiDAR Feature Learning via a Morphological Network

Mathematical morphology is known to be powerful for analyzing the intrinsic shape, structure, and size of the texture object in the image. Here, a morphological network is designed based on two primitive operations, i.e., dilation and erosion using the widely used SEs of size $(s \times s)$ kernel. The channel of LiDAR can be thought of as a surface over the high-dimensional plane.

The input LiDAR patches are convolved with such kernels to produce the dilated image, which contains maximum overall pixels in its neighborhood defined locally. The dilation operation increases the boundaries of foreground pixels regions on the input LiDAR image. Thus, the texture of various regions in the LiDAR is enlarged depending on the size of the kernel.

Similarly, the output of the erosion operation contains the minimum overall pixels of its local neighborhood during the convolution with the kernel. Opposite to the dilation operation,

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erosion reduces the shape of the texture region in the LiDAR image. It is worth noting that the erosion eliminates the minor details, enlarges holes, and makes them separable between different texture regions. Let $X_L \in R^{k\times k}$ be the input single channel LiDAR patches of spatial size ($k \times k$). We note that the spatial sizes of both HSIs and LiDAR patches are kept the same. The dilation and erosion operations can be denoted by $\oplus$ and $\ominus$, respectively, since both the dilation and erosion operations are nonlinear and produce the output by interacting between the whole LiDAR image X_L and a set called SEs, which can be defined as follows:

$$(X_L \oplus W_d)(x, y) = \max_{(i,j)\in\psi} (X_L(x+i, y+j) + W_d(i, j)) \quad (2)$$

$$(X_L \ominus W_e)(x, y) = \min_{(i,j)\in\psi} (X_L(x+i, y+j) - W_e(i, j)) \quad (3)$$

where $\psi = \{(i, j) \mid i \in \{1, 2, 3, \ldots, s\};\ j \in \{1, 2, 3, \ldots, s\}\}$ represent the elements of the kernel, and W_d and W_e are the kernels or SEs for the dilation and erosion operation, respectively. It can be noted that the operations defined in (2) and (3) are ($s \times s$) window specific and closely similar with the convolutional layers. To keep the input and output shape same, the padding function is used after the dilation and erosion operation. After applying (2) and (3) over the LiDAR patch, the results are obtained as dilation map and erosion map, respectively.

Fig. 3(a) and (b) describes the input and output images after the dilation and erosion operations using defined SEs of size $3 \times 3 \times 1$. The dilation will enlarge the bright regions while shrinking the dark ones of a particular texture feature in the morphological space. Similarly, erosion will suppress the texture feature of SEs size. To obtain the morphological shape feature from LiDAR data, a spatial morphological (`SpatialMorph`) block (as shown in Fig. 2) using the primitive operations is introduced in this article. The proposed `SpatialMorph` block $\mathcal{F}_{\text{Morph}}(X_L, W_d, W_e, \theta)$ is derived using the (2) and (3) and given as follows:

$$\mathcal{F}_{\text{Morph}}(X_L, W_d, W_e, \theta) = F_{2D}^{(1,1)}((X_L \oplus W_d), \theta) \odot F_{2D}^{(1,1)}((X_L \ominus W_e), \theta) \quad (4)$$

where W_d and W_e are the weights of the (3×3) kernel and $F_{2D}^{(1,1)}$ is the function that denotes the linear combination between dilation and erosion feature maps obtained using 2-D convolution parameterized with θ and kernels of size (1×1). To obtain the resulting feature maps, a linear combination of both dilation and erosion LiDAR features is calculated as follows:

$$F_{2D}^{(1,1)}\left(X^{\text{int}}\right) = b + \sum_{l=1}^{m} w_l X^{\text{int}}(x, y, l) \quad (5)$$

where X^{int} represents the intermediate dilation or erosion feature maps, w_l represents the weights for the lth feature channel, and b represents the bias of the network. The SEs and weights of the linear combination are initialized randomly.

C. Hyperspectral and LiDAR Feature Fusion

The spectral–spatial feature maps are extracted from X_H using 3-D CNNs. Similarly, the spatial and complementary feature maps are also obtained from X_L using morphological CNN. To better learn both the complementary feature during training of the network and the fusion of both the features became an important issue for HSIs and LiDAR classification.

The traditional way to combine features from a different source is to stack them and fuse the feature using the FC layers. Due to the large size of the combined feature vector, the linear layer often suffers from huge parameter phenomena, which creates difficulty in training. To balance both the issues, here, we use simple channelwise feature concatenation of both the feature maps R_H and R_L immediately after the adaptive average pooling layers

$$O_{\text{HL}} = \mathcal{F}_{\text{pool}}[\mathcal{F}_{3D}(X_H, \theta_1)] \odot \mathcal{F}_{\text{pool}}\left[\mathcal{F}_{\text{Morph}}(X_L, \theta_2)\right] \quad (6)$$

where $O_{\text{HL}} \in \mathcal{R}^{64\times1\times1}$ represent the fused output of the obtained feature maps $R_H \in \mathcal{R}^{32\times1\times1}$ and $R_L \in \mathcal{R}^{32\times1\times1}$. $\mathcal{F}_{\text{pool}}$ is adaptive average ppooling operation, whereas X_H and X_L are the input HSI and LiDAR data for the two baseline networks, i.e., 3-D CNN ($\mathcal{F}_{3D}$) and spatial morphological CNN ($\mathcal{F}_{\text{Morph}}$), which are parameterized with θ_1 and θ_2, respectively.

Later, the obtained fused feature maps O_{HL} are passed through a flatten layer to produce a feature vector, which is followed by a linear layer for accurate prediction of land-cover classes. The output value of $\hat{y}$ can be derived as follows:

$$\hat{y} = \mathcal{F}_{\text{flatten}}(O_{\text{HL}}, \omega) = \mathcal{F}_{\text{linear}}(\omega O_{\text{HL}}) \quad (7)$$

where $\hat{y} \in \mathcal{R}^{C\times1}$ is the prediction, C is the number of land-cover classes for the corresponding datasets, and $\mathcal{F}_{\text{flatten}}$ is the linear layer parameterized with ω.

D. Training and Testing of the Joint Network

The proposed model (as shown in Fig. 2) is trained using an end-to-end strategy where D^{train} and D^{test} represent the training and testing sets, respectively. The training and test sets contain P and Q numbers of labeled and unlabeled land-cover samples for the training and validation of the proposed network. The network is trained using the Adam [58], [59] optimizer and backpropagates the training loss. The training cross-entropy loss L_{CE} between the predicted output $\hat{y}^{(i)}$ and the targeted output $y^{(i)}$ can be calculated as follows:

$$L_{\text{CE}} = \frac{1}{N}\sum_{i=1}^{N} y^{(i)} \log\left(\hat{y}^{(i)}\right) + \left(1 - y^{(i)}\right) \log\left(1 - \hat{y}^{(i)}\right). \quad (8)$$

IV. Experimental Results and Discussion

A. Hyperspectral Datasets

To evaluate the performance of the proposed and comparative networks, three different HSI scenes along with their associated LiDAR data have been considered. The experimental datasets include the University of Houston (UH), Trento, and MUUFL Gulfport (MUUFL) scenes. All these datasets are further described in the following.

1) The IEEE Geoscience and Remote Sensing Society published the University of Houston (UH) dataset—collected by the Compact Airborne Spectrographic

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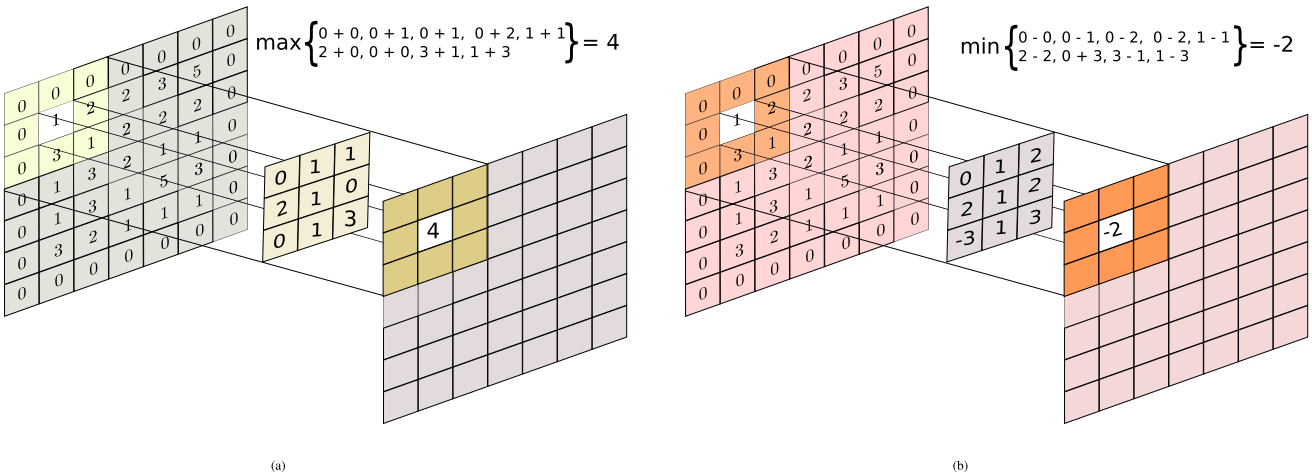
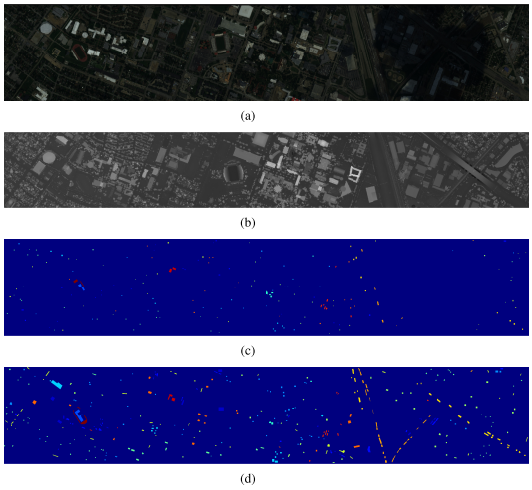


Fig. 3. Graphical visualization of the dilation and erosion operations where an input image patch of size ($7 \times 7 \times 1$) is dilated and eroded with an SE of size ($3 \times 3 \times 1$), and the obtained output size keeps the same with a padding mechanism. (a) Dilation Operation. (b) Erosion Operation.

Imager (CASI) in 2013, as part of its Data Fusion Contest. The datasets consist of HSIs and LiDAR data, and both are composed of 340 × 1905 pixels with 144 spectral bands. The spatial resolution of this dataset is 2.5 MPP with a wavelength ranging from 0.38–1.05 μm. Finally, the ground truth comprises 15 different land-cover classes. In addition, the available samples are divided into disjoint training and test samples of 15 classes. The lists of disjoint training and test samples for each of the 15 different land-cover classes are shown in Fig. 4.

2) The MUUFL Gulfport scene was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the campus of University of Southern Mississippi Gulf Park, Long Beach Mississippi, in November 2010 [60], [61]. It consists of 325 × 220 pixels with 72 spectral bands, and the LiDAR modality consists of elevation of two raster data. However, due to noise, the initial and final eight bands are omitted leading to a total of 64 bands. There are a total of 53 687 ground-truth pixels describing 11 urban land-cover classes. Fig. 5 shows the list of 5% randomly chosen samples from every 11 different classes.
3) The Trento scene was gathered by using the AISA eagle sensor over the rural regions in the south of Trento, Italy, while the LiDAR data were captured by the Optech ALTM 3100EA sensor. The HSI contains 63 spectral bands within a wavelength of range 0.42–0.99 μm, and the LiDAR data has two rasters exploring elevation data. The scene has 600 × 166 pixels that comprise six mutually exclusive vegetation land-cover classes where the spectral resolution is 9.2 nm and the spatial resolution is 1 MPP. In addition, the available samples are divided into disjoint training and test samples of six classes. Fig. 6 lists the information about the per class number of samples in six different land-cover classes.

Figs. 4–6 show a detailed summary of the UH, MUUFL, and Trento scenes, respectively, including their corresponding ground truth, the type associated with the land-cover classes, and the number of available labeled samples per class.



Color	Land cover	Train	Test	Color	Land cover	Train	Test
	Background	662013	652648		Grass-healthy	198	1053
	Grass-stressed	190	1064		Grass-synthetic	192	505
	Tree	188	1056		Soil	186	1056
	Water	182	143		Residential	196	1072
	Commercial	191	1053		Road	193	1059
	Highway	191	1036		Railway	181	1054
	Parking-lot1	192	1041		Parking-lot2	184	285
	Tennis-court	181	247		Running-track	187	473

F 4 V l f h U f H (UH) ( ) P d l f

Fig. 4. Visualization of the University of Houston (UH) scene. (a) Pseudocolor image for the HSI data on bands 64, 43, and 22, respectively. (b) Grayscale image for the LiDAR data. (c) Ground truth of disjoint training samples. (d) Ground truth of disjoint test samples. The table represents class-specific land-cover types and the number of disjoint training and test samples i.e., 664 845 and 664 845, respectively. (a) HSI pseudocolor map. (b) LiDAR map. (c) Disjoint training samples. (d) Disjoint test sample.

B. Experimental Protocols

In order to explore the effectiveness of the proposed network, several state-of-the-art methods have been comprehensively studied and compared. The comparative methods

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Color	Land cover	Train	Test	Color	Land cover	Train	Test
	Background	68816	20497		Trees	1162	22084
	Grass-Pure	214	4056		Grass-Groundsurface	344	6538
	Dirt-And-Sand	91	1735		Road-Materials	334	6353
	Water	23	443		Buildings'-Shadow	112	2121
	Buildings	312	5928		Sidewalk	69	1316
	Yellow-Curb	9	174		ClothPanels	14	255

Fig. 5. Visualization of the MUUFL scene. (a) True-color image for the HSI data over bands 40, 20, and 10, respectively. (b) Grayscale image for the LiDAR data. (c) Ground truth of MUUFL scene. The table represents class-specific land-cover types and the number of randomly selected 5% training and remaining 95% test samples, i.e., 71 500 and 71 500, respectively. (a) HSI pseudocolor map. (b) LiDAR map. (c) Ground-truth map.

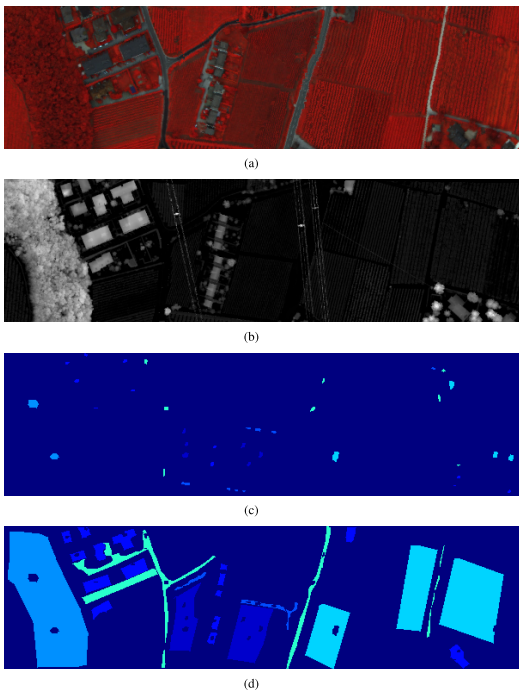
include both classical models and deep learning-based feature fusion models. The comparative methods include random forest (RF) [62], SVM with radial basis function [15], the recurrent neural network (RNN) [63], Two-CNN [50], FusAtNet [64], CoupleNet [8], HWRN [12], HybridSN [33], EndNet [65], and the proposed network.

The performance of the proposed network along with the comparative methods is evaluated in terms of several widely used quantitative measurements, such as overall accuracy (OA), average accuracy (AA), and statistical Kappa (κ) coefficient, respectively. OA is referred to the ratio of correctly classified samples among the total test samples, whereas the AA represents the mean of classwise accuracy. Moreover, κ represents a strong mutual agreement between the generated classification maps of the considered model and the provided ground truth.

For experimental validation and to prove the superiority of our proposed network, a five-cross-validation process has been adopted and executed five times using 200 epochs per iteration. The reported accuracies are the average result of an aforesaid process. All the tests have been performed on a Red Hat Enterprise Server (Release 7.6) with a CPU having ppc64le architecture and a total of 40 cores with four threads per core and 377 GB of RAM. The GPU used is a single Nvidia Tesla V100 with 32 510 MB of VRAM. The source code of the proposed MFT was implemented using PyTorch 1.5.0 and Python 3.7.7.

Moreover, the experimental process has been conducted in three different settings.

1) Experiments have been conducted on disjoint (spatially and spectrally) training and test samples, i.e., the intersection between the training samples and test samples remains empty.
2) Different percentages of training samples have been used to validate the performance of our proposed network.
3) Different spatial dimensions have also been considered for experimental validation.



Color	Land cover	Train	Test	Color	Land cover	Train	Test
	Background	98781	70205		Apples	129	3905
	Buildings	125	2778		Ground	105	374
	Woods	154	8969		Vineyard	184	10317
	Roads	122	3052				

Fig. 6. Visualization of the Trento data. (a) True color image for the HSI data over bands 40, 20, and 10, respectively. (b) Grayscale image for the LiDAR data. (c) Ground truth of disjoint training samples. (d) Ground truth of disjoint test samples. The table represents class-specific land-cover types and the number of disjoint training and test samples, i.e., 99 600 and 99 600, respectively. (a) HSI pseudocolor map. (b) LiDAR map. (c) Disjoint training samples. (d) Disjoint test sample.

C. Performance Evaluation With Disjoint Train/Test Samples

In order to verify the performance of our proposed network, a qualitative and quantitative comparison has been conducted between the proposed method and several state-of-the-art methods mentioned above. For a fair analysis and to make the results reliable, each experiment has been repeated five times; both the proposed and comparative methods have been evaluated in the same experimental settings. The experimental protocols are set as follows for the disjoint train/test samples experiment. To process the HSI image, the spatial patch size is set to ($11 \times 11 \times B$), where B is the number of bands for, respectively, datasets, and the number of training epochs is set to 200 without data augmentation and BN. The batch size is set to 32, and the weights are optimized using the Adam optimizer. Moreover, the learning rate of each method is set as mentioned in their respective papers, whereas the learning rate of our proposed method is set to 0.001. Finally, the training and test samples' percentages are set to 5%, i.e., 5% randomly selected training samples, and the rest of the 95% disjoint (spatially/spectrally) samples are unseen samples for model evaluation.

The first kind of experiment is processed to check the artifacts of disjoint training/test samples on different models, including the one proposed in this article. The disjoint

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TABLE I

OA, AA, AND KAPPA VALUES ON THE UNIVERSITY OF HOUSTON DATASET (IN %). "H" REPRESENTS ONLY HSI, WHILE "H + L" REPRESENTS FUSED HSI AND LiDAR

Class	RF [1]		SVM [15]		RNN [63]		Two-CNN [50]		FusAtNet [64]		CoupledNet [8]		HWRN [12]		HybridSN [33]		EndNet [65]		Proposed	
	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L
1	82.90	82.90	81.48	81.38	81.76	82.24	81.86	83.09	81.38	81.38	82.05	82.14	81.29	82.62	82.43	82.52	76.44	79.83	**83.00**	**93.54**
2	82.42	81.95	78.94	81.58	76.22	84.49	84.39	84.39	84.39	82.80	82.04	72.08	79.98	84.86	83.08	82.89	83.61	77.16	**84.86**	**84.77**
3	64.15	64.75	56.83	59.00	82.97	71.09	63.36	96.43	71.88	82.57	60.19	84.75	88.51	80.19	81.18	96.23	96.36	99.33	**95.64**	**98.81**
4	91.85	92.04	84.09	86.08	97.72	91.38	91.38	93.27	91.38	91.19	89.29	93.27	86.93	92.80	92.70	99.90	73.57	80.11	**97.91**	**95.07**
5	99.81	99.90	94.50	95.07	99.24	97.16	95.92	98.95	99.90	100.0	95.73	96.87	96.68	96.87	99.05	98.29	98.04	99.43	**99.90**	**99.52**
6	97.20	97.90	72.02	78.32	67.83	86.71	79.72	97.20	94.40	95.10	86.71	79.72	89.51	86.01	92.30	90.90	95.10	95.33	**95.80**	**95.80**
7	86.94	88.15	82.09	85.63	80.31	82.37	81.52	89.27	78.17	88.33	84.32	83.48	73.04	87.77	86.38	83.30	65.20	78.17	**88.61**	**93.00**
8	41.78	75.40	47.57	47.67	41.88	61.25	66.66	86.70	64.38	72.65	70.18	82.14	75.97	75.97	72.74	67.52	79.74	88.69	**76.06**	**86.70**
9	**85.74**	**90.84**	78.37	82.90	79.13	77.90	79.32	77.43	81.02	84.70	78.94	88.66	78.47	86.68	74.50	80.83	81.24	80.95	82.24	84.60
10	35.90	38.99	46.52	48.84	39.67	41.31	58.59	69.69	57.72	64.38	52.41	52.31	49.22	51.06	**58.49**	42.08	60.71	60.52	48.35	**67.18**
11	64.51	73.62	47.43	56.83	53.22	55.31	62.71	85.48	63.37	79.79	66.41	74.47	66.22	67.83	73.43	64.23	76.56	83.27	**84.72**	**99.62**
12	67.81	69.54	65.99	75.40	84.43	85.01	92.79	93.75	91.06	81.84	81.94	81.55	80.21	84.82	85.68	83.09	93.08	95.16	**97.02**	**98.17**
13	88.77	87.01	74.73	80.00	50.87	68.07	90.17	91.22	84.21	**92.98**	80.70	80.00	84.91	82.10	**92.98**	89.12	73.68	76.14	92.28	92.63
14	100.0	100.0	72.47	72.47	91.09	93.52	76.51	86.64	93.11	77.32	78.13	86.64	90.68	86.23	88.26	86.64	89.33	96.22	**100.0**	**100.0**
15	90.90	99.78	48.20	52.22	88.79	75.26	63.00	81.39	81.60	73.78	74.20	71.24	99.57	97.46	75.26	99.78	95.48	98.73	**100.0**	**100.0**
OA	75.47	80.43	69.47	72.88	74.19	75.99	78.43	86.71	79.66	82.67	77.62	80.72	78.83	81.97	81.25	80.66	80.47	83.97	**86.04**	**91.30**
AA	78.71	82.85	68.75	72.23	74.34	76.87	77.86	87.66	81.20	83.25	77.55	80.69	81.41	82.89	82.56	83.15	82.54	85.93	**88.43**	**92.63**
$\kappa(\times 100)$	73.59	78.93	67.02	70.73	72.05	74.03	76.66	85.59	78.06	81.26	75.78	79.15	77.08	80.50	79.70	79.00	78.86	82.63	**84.88**	**90.55**

TABLE II

OA, AA, AND KAPPA VALUES ON THE UNIVERSITY OF TRENTO DATASET (IN %). "H" REPRESENTS ONLY HSI, WHILE "H + L" REPRESENTS FUSED HSI AND LiDAR

Class	RF [1]		SVM [15]		RNN [63]		Two-CNN [50]		FusAtNet [64]		CoupledNet [8]		HWRN [12]		HybridSN [33]		EndNet [65]		Proposed	
	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L
1	97.49	97.18	95.90	96.00	96.18	95.87	97.87	98.74	96.72	**99.59**	98.97	98.20	98.87	98.18	95.82	98.95	93.52	93.06	**100.0**	99.23
2	89.63	98.88	80.31	82.25	89.30	86.61	91.93	**98.23**	87.90	92.80	92.62	95.96	**94.60**	95.03	63.93	82.29	93.76	99.42	91.54	97.76
3	75.93	71.65	63.63	61.49	50.53	60.96	78.87	78.61	53.47	66.31	71.12	69.51	65.77	63.36	74.06	65.77	87.43	88.23	**94.65**	**97.32**
4	99.96	**100.0**	90.03	92.78	99.45	99.91	97.93	99.64	98.08	95.85	98.81	99.93	98.30	99.16	99.66	99.72	98.74	99.69	**99.98**	**100.0**
5	99.96	99.99	99.41	99.38	99.03	96.83	99.94	**100.0**	97.80	97.13	99.88	99.56	99.89	99.76	98.58	99.67	98.21	99.23	**99.98**	**100.0**
6	67.00	90.26	87.22	95.80	60.19	71.98	64.64	88.04	60.25	77.22	65.79	88.89	65.26	84.60	90.62	75.36	86.74	87.10	**90.69**	**94.07**
OA	94.93	98.14	92.56	94.44	93.21	93.64	94.36	98.04	92.34	94.20	94.84	97.66	94.74	96.88	94.13	94.99	96.00	97.17	**98.15**	**99.04**
AA	88.33	92.99	86.08	87.95	82.45	85.36	88.53	93.87	82.37	88.15	87.87	92.01	87.12	90.02	87.11	86.96	93.06	94.46	**96.14**	**98.06**
$\kappa(\times 100)$	93.20	97.51	90.08	92.59	90.90	91.49	92.45	97.37	89.66	92.23	93.09	96.87	92.95	95.81	92.12	93.27	94.65	96.21	**97.52**	**98.71**

TABLE III

OA, AA, AND KAPPA VALUES ON THE MUUFL GULFPORT DATASET (IN %). "H" REPRESENTS ONLY HSI, WHILE "H + L" REPRESENTS FUSED HSI AND LiDAR OVER 5% TRAINING SAMPLES

Class	RF [1]		SVM [15]		RNN [63]		Two-CNN [50]		FusAtNet [64]		CoupledNet [8]		HWRN [12]		HybridSN [33]		EndNet [65]		Proposed	
	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L
1	98.08	98.09	98.08	98.35	96.07	96.43	97.33	97.58	98.74	98.15	97.68	97.88	98.33	98.16	97.12	98.12	89.03	91.82	**98.37**	**97.46**
2	76.50	77.19	57.32	55.52	80.69	79.29	82.27	82.29	79.78	80.32	76.15	83.50	79.51	74.26	86.16	82.27	77.11	84.61	**90.08**	**90.38**
3	85.89	85.71	73.87	72.89	85.25	86.25	86.12	91.20	86.67	89.08	86.12	86.92	88.22	88.36	85.28	**89.79**	78.66	74.13	**90.74**	89.72
4	83.97	83.91	48.53	39.53	87.55	88.87	90.83	92.10	85.93	88.24	91.93	92.10	94.00	93.31	87.14	93.37	77.80	86.39	**93.83**	**96.13**
5	92.82	93.23	81.45	80.12	88.00	90.10	92.72	95.57	93.90	94.23	94.25	92.47	**95.68**	94.27	92.08	92.79	79.03	82.15	94.49	**95.94**
6	89.16	88.71	28.44	30.92	78.78	82.84	81.71	91.19	75.62	89.84	65.46	95.26	74.04	88.48	75.62	94.35	90.14	96.61	**97.06**	**97.97**
7	82.60	82.69	58.65	54.40	85.05	85.66	87.88	90.66	85.80	87.88	87.74	85.57	89.67	90.19	**90.19**	91.65	77.03	82.39	86.23	**92.92**
8	90.46	90.38	85.96	86.37	87.45	87.11	96.28	96.13	93.25	95.85	96.76	**97.23**	**97.42**	96.96	92.51	92.45	93.32	94.97	96.92	96.23
9	44.83	44.83	25.45	18.61	59.49	61.62	**65.27**	56.00	55.92	58.20	57.14	57.52	60.79	**63.98**	60.56	59.27	30.64	50.78	30.09	39.74
10	13.21	16.66	7.47	07.47	28.73	**42.52**	**45.40**	00.00	09.77	27.58	18.96	17.81	39.65	37.35	29.88	25.28	27.20	31.22	00.00	00.00
11	55.29	55.29	00.00	00.00	79.60	55.29	65.49	68.62	80.78	85.49	75.68	74.11	**91.76**	**89.41**	62.35	65.88	76.82	86.79	65.88	82.35
OA	90.18	90.27	81.65	80.74	89.30	89.78	92.09	93.05	91.77	92.64	91.64	92.42	93.18	92.66	91.41	92.59	83.36	86.31	**93.16**	**93.45**
AA	73.89	74.25	51.38	49.47	77.88	77.82	81.03	78.30	76.92	81.35	77.08	80.03	**82.64**	**83.16**	78.08	80.47	72.43	78.35	76.70	79.89
$\kappa(\times 100)$	86.88	87.01	74.33	72.89	85.86	86.47	89.53	90.79	88.97	90.18	88.89	89.93	90.94	90.25	88.62	90.19	78.36	82.17	**90.90**	**91.33**

train/test samples significantly overcome the redundancy among the seen and unseen samples, which is a prerequisite condition of fair evaluation of any model. Usually, the remote sensing researcher tends to use the randomly selected training samples to build the model; however, the entire HSI datasets are used at the testing time, which induces a strong bias. Such a model does produce a higher accuracy but lacks confidence especially when it comes to evaluating unseen samples. Moreover, disjoint train/test samples also help to avoid the overlapping region issues of HSI, especially at the training time, i.e., the model has been evaluated on spectrally and spatially disjoint samples, thus bringing more confidence.

The experimental results in terms of OA, AA, kappa (κ), and classwise accuracies on disjoint train/test samples for the University of Houston and Trento datasets are presented in Tables I–II. On the other hand, Table III presents the results on 5% randomly selected training samples (with the rest of the samples used for testing) for the MUUFL dataset as geographically disjoint train/test samples for MUUFL are not available to the research community. Tables I–III show results for both the HSI dataset and the fused HSI and LiDAR datasets. The "H" column represents the results for the HSI datasets (in the case of the proposed method, the LiDAR branch is turned off), while the "H + L" column shows the results for the fused HSI and LiDAR datasets (in the case of the proposed method, both HSI branch and LiDAR branch are turned on). It is evident that the overall performance of almost all models improves with the use of LiDAR in conjunction with HSI data compared to using HSI data alone. Figs. 7–9 illustrate the classification maps for the University of Houston, Trento, and MUUFL datasets, respectively. It can be clearly seen that the classification maps for the proposed model contain less salt and pepper noise compared to the classification maps of other models. From these results, one can conclude that the proposed method (in both scenarios) works better in almost 5% OA for HSI and 11% for HSI + LiDAR case compared to the HybridSN; 7% for HSI and 10% for HSI + LiDAR compared to HWRN; 9% for HSI and 11%

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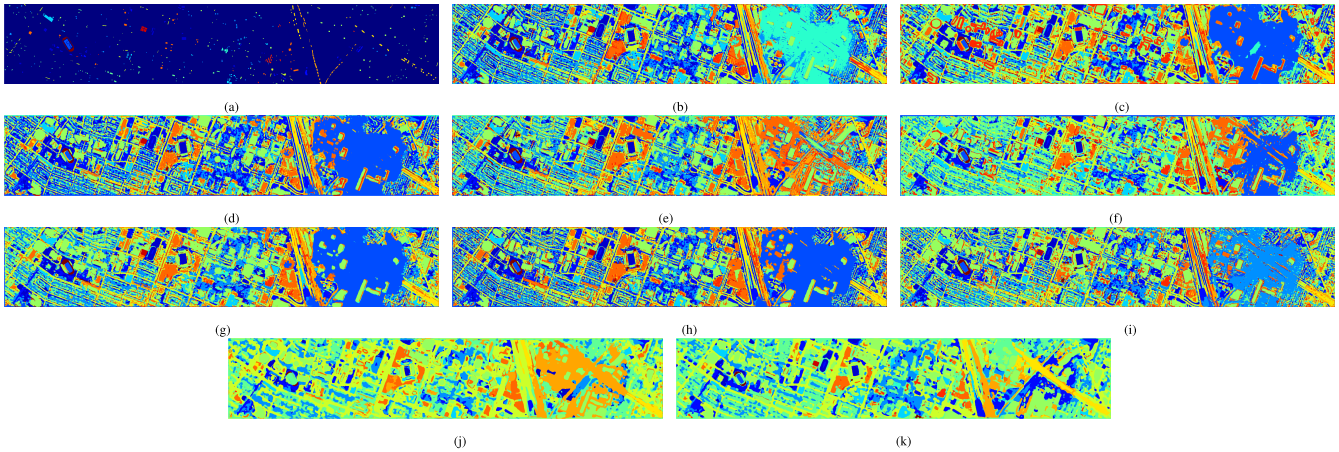


Fig. 7. (a) False color representation of the first PC obtained from the UH scene. Ground-truth classification maps along with the kappa (κ) accuracy obtained for the DUH dataset for both HSI and combined HSI and LiDAR, (b) RF = 73.59 (HSI) and 78.93 (HSI + L), (c) SVM = = 67.02 (HSI) and 70.73 (HSI + L), (d) RNN = 72.05 (HSI) and 74.03 (HSI + L), (e) Two-CNN = 76.66 (HSI) and 85.59 (HSI + L), (f) FusAtNet = 78.06 (HSI) and 81.26 (HSI + L), (g) CoupledNet = 75.78 (HSI) and 79.78 (HSI + L), (h) HWRN = 77.08 (HSI) and 80.50 (HSI + L), (i) HybridSN = 79.70 (HSI) and 79.00 (HSI + L), (j) EndNet = 78.86 (HSI) and 82.63 (HSI + L), and (k) proposed model = **84.88** (HSI) and **90.55** (HSI + L). The higher accuracies are in bold face.

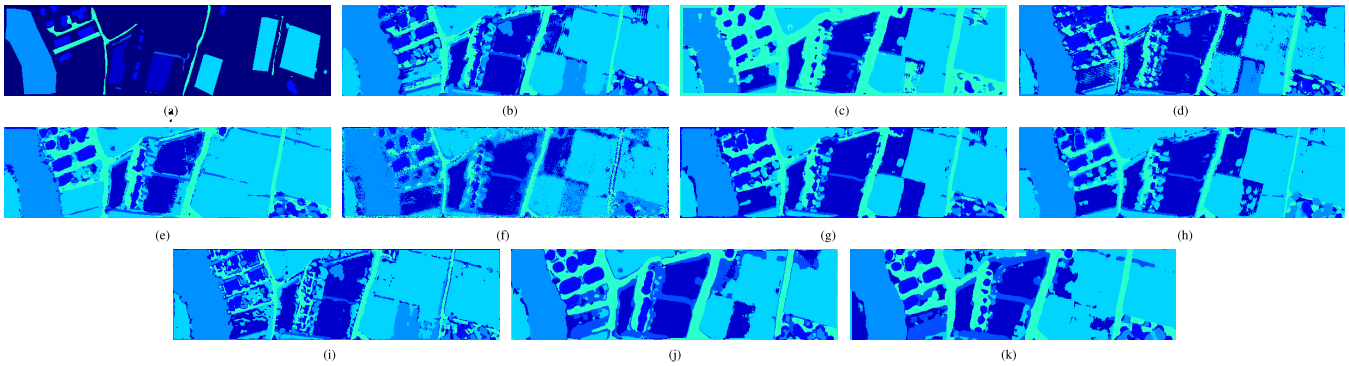


Fig. 8. (a) False color representation of the first PC obtained from the Trento scene. Ground-truth (GT) classification maps along with the kappa (κ) accuracy obtained for the DUH dataset for both HSI and combined HSI and LiDAR. (b) RF = 93.20 (HSI) and 97.51 (HSI + L), (c) SVM = 90.08 (HSI) and 92.59 (HSI + L), (d) RNN = 90.90 (HSI) and 91.49 (HSI + L), (e) Two-CNN = = 92.45 (HSI) and 97.37 (HSI + L), (f) FusAtNet = 89.66 (HSI) and 92.23 (HSI + L), (g) CoupledNet = 93.09 (HSI) and 96.87 (HSI + L), (h) HWRN = 92.95 (HSI) and 95.81 (HSI + L), (i) HybridSN = 92.12 (HSI) and 93.27 (HSI + L), (j) EndNet = 94.65 (HSI) and 96.21 (HSI + L), and (k) proposed model = **97.52** (HSI) and **98.71** (HSI + L). The higher accuracies are in bold face.

for HSI + LiDAR compared to CoupledNet; 6% for HSI and 9% for HSI + LiDAR compared to FusAtNet; 4% for HSI and 5% for HSI + LiDAR compared to Two-CNN; 8% for HSI and 16% for HSI + LiDAR compared to RNN; 17% for HSI and 20% for HSI + LiDAR compared to SVM; and 13% for HSI and 12% for HSI + LiDAR compared to RF. Though some methods achieve superior classwise accuracies using just HSI in some cases, the overall performance using fused HSI and LiDAR is far better. Similar observations can be made for the other experimental datasets in which the proposed network outperformed state-of-the-art and conventional methods.

It has been observed that both convolution, and morphological dilation and erosion are the neighborhood operators with respect to the trainable kernels [66], [67]. To extract the features from HSI data, the convolutional layer uses a set of trainable filters that allow performing linear combinations between the HSI cubes and the weights of the kernel. However, it is not enough for HSI data to distinguish between two different classes by considering only the spectral information since both the classes produce similar spectral responses. To provide complementary information to the HSI data, LiDAR data were considered, which successfully captured the elevation information of each class in an image, such as height, width, and shape, covering the same surveyed area. Unlike convolution operations, morphological operators are nonlinear operations in nature that can better approximate the shape information of the image classes.

To achieve this, the morphological layer (i.e., dilation and erosion) replaces the linear convolution operation with trainable morphological max or min operators, which allows the proposed network to capture distinct and robust elevation information from LiDAR data compared to conventional CNNs. Both the dilation and erosion feature maps preserve the nonlinear structure of the LiDAR data and are parallelly learned to aggregate the most important nonlinear information from LiDAR data, and both the feature maps are normalized using a 2-D convolution operation with a kernel of size (1×1), which reduces the feature discrepancy obtained from dilation and erosion layers.

However, the LiDAR data also fail to discriminate between two different roads even though roads are formed using different materials, but both roads produced similar elevation information. Therefore, it is a good choice to combine both HSI and LiDAR information, which provides complementary

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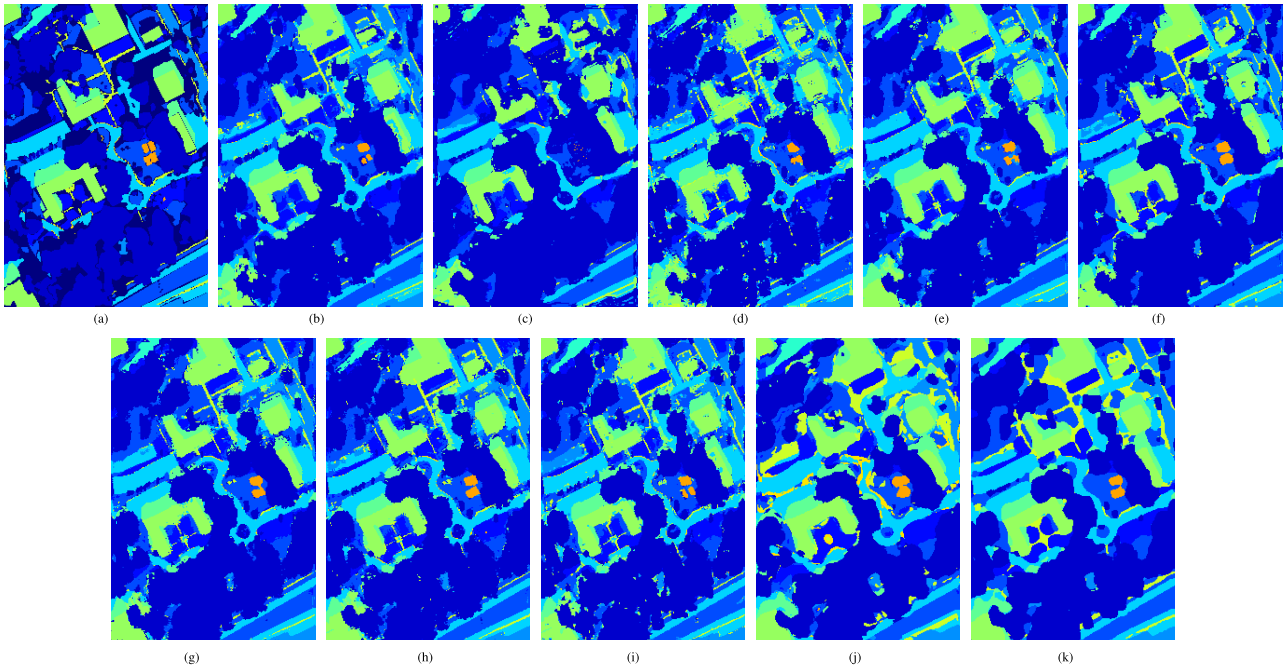


Fig. 9. (a) False color representation of the first PC obtained from the MUUFL scene. Ground-truth (GT) classification maps along with the kappa (κ) accuracy obtained for the DUH dataset for both HSI and combined HSI and LiDAR. (b) RF = 86.88 (HSI) and 87.01 (HSI + L), (c) SVM = 74.33 (HSI) and 72.89 (HSI + L), (d) RNN = 85.86 (HSI) and 86.47 (HSI + L), (e) Two-CNN = 89.53 (HSI) and 90.79 (HSI + L), (f) FusAtNet = 88.97 (HSI) and 90.18 (HSI + L), (g) CoupledNet = 88.89 (HSI) and 89.93 (HSI + L), (h) HWRN = 90.94 (HSI) and 90.25 (HSI + L), (i) HybridSN = 88.62 (HSI) and 90.19 (HSI + L), (j) EndNet = 78.36 (HSI) and 82.17 (HSI + L), and (k) proposed model = **90.90** (HSI) and **91.33** (HSI + L). The higher accuracies are in bold faces.

information to each other and can significantly improve classification performance. In the above context, we employed morphological dilation and erosion layers for their usefulness to learn nonlinear object boundaries for complex classes from LiDAR data. The fused information of HSI and LiDAR outperformed the existing results in terms of OA, AA, and Kappa coefficient.

D. Performance Evaluation With Different Spatial Dimensions

The classification performance of any spectral–spatial network is also dependent on the spatial dimension of the input cubes. Thus, the patch size has an important impact on the result. Therefore, this section presents several experimental results with different patch sizes to process HSI cubes to explore the impact of different patch sizes on classification results. In all these experiments, the initial training/test samples percentages are set as explained in the previous sections. All other training parameters remain the same except the patch sizes that are tested as 7×7, 9×9, and 11×11. The OA, AA, and κ accuracies have been considered and presented in Table IV to validate the performance of the proposed network along with the comparative methods on different patch sizes. The optimal results are highlighted in bold across all the spatial sizes per dataset.

The superiority of the proposed model across various spatial windows is also quite evident from Table IV. The proposed model achieves 97.29% and 99.70% accuracies both over the 11×11 and 94.16% over 7×7 in terms of OAs across all the windows for UH, Trento, and MUUFL datasets respectively. Similarly, AAs of 97.35% and 99.31% are achieved by the proposed network using 11×11 and 7×7, respectively, for both the UH and Trento datasets, whereas FusAtNet shows its prowess in the MUUFL dataset over the 11×11 window. Finally, the proposed network achieves 97.07% and 99.61% of Kappa values for UH and Trento datasets over the window of size 11×11 and 92.27% for the MUUFL dataset over 9×9, and dominates all other models. HWRN can only achieve better AA (83.12%) than the proposed model for the MUUFL dataset across all the spatial windows. All deep learning methods (Two-CNN, FusAtNet, CoupledNet, HWRN, HybridSN, and EndNet) perform competitively but cannot achieve the results provided by the proposed model.

From Table IV, one can find that the classification performance significantly improves as the spatial patch size increases. As a matter of fact, the larger the patch size, the most information can be characterized, thus producing better accuracies. The fact remains the same for all experimental datasets until the patch contains the interfering samples. Since, if some interfering samples appeared in the same patch, then the accuracies may decrease significantly. Therefore, a suitable patch size is very crucial for appropriate classification results.

E. Performance Evaluation With Different Numbers of Training Samples

Despite the aforesaid experiments, there is another challenge that needs to be evaluated to validate the claims, i.e., the number of labeled training samples. If the percentage of the training samples is not adequate or not reliable, it may lead to the under-fitting (also known as the Hughes phenomena) or overfitting (not reliable) issues. Thus, how to select an appropriate number of training samples is another important factor for classification accuracy. This section presents rigorous experimental results on different percentages of training

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TABLE IV

OVERALL (OA), AVERAGE (AA), AND KAPPA (κ) ACCURACIES (%) OF RF, SVM, RNN, TWO-CNN, FusAtNet, CoupledNet, HWRN, HybridSN, AND THE PROPOSED NETWORK USING DIFFERENT SPATIAL WINDOW SIZES, I.E., 7×7, 9×9, AND 11×11, RESPECTIVELY

Datasets	Spatial Window	RF [1]	SVM [15]	RNN [63]	Two-CNN [50]	FusAtNet [64]	CoupledNet [8]	HWRN [12]	HybridSN [33]	EndNet [65]	Proposed
		Overall Accuracy (OA)									
UH	7x7	89.71	76.32	92.26	89.49	86.03	93.08	92.28	93.76	93.51	95.98
Trento		98.87	86.98	98.92	99.54	98.88	99.48	99.60	99.37	99.43	99.63
MUUFL		89.44	88.04	90.30	90.56	89.23	91.45	91.95	88.21	88.53	93.71
UH	9x9	90.45	68.69	92.52	91.75	84.65	93.31	92.17	93.71	94.02	97.04
Trento		98.85	78.10	99.13	99.17	99.36	99.54	99.62	99.21	99.22	99.68
MUUFL		89.96	85.71	89.75	91.78	91.51	92.12	91.72	88.57	86.73	**94.16**
UH	11x11	91.19	61.55	91.22	90.79	87.75	94.02	93.26	94.27	93.65	**97.29**
Trento		98.80	72.66	99.09	99.45	99.46	99.23	99.55	99.34	98.93	**99.70**
MUUFL		90.25	80.74	89.90	93.19	91.76	92.54	92.84	91.98	85.89	93.31
		Average Accuracy (AA)									
UH	7x7	88.81	73.94	90.51	88.22	82.83	91.83	91.55	93.08	93.55	96.03
Trento		97.67	78.49	97.56	98.45	95.52	98.45	98.92	98.59	98.82	**99.31**
MUUFL		75.02	70.89	81.18	74.95	73.28	78.48	**83.12**	76.94	80.47	82.50
UH	9x9	89.42	65.85	91.20	90.75	81.59	91.34	91.16	93.00	93.86	97.02
Trento		97.58	64.91	98.10	94.48	98.23	98.14	98.56	97.69	97.95	99.05
MUUFL		74.60	60.16	77.67	75.95	77.78	78.66	81.48	76.27	78.91	80.59
UH	11x11	90.04	57.31	89.50	87.97	84.66	92.00	92.26	94.47	93.11	**97.35**
Trento		97.45	55.30	98.15	97.37	98.67	96.74	98.26	98.24	97.52	99.16
MUUFL		74.04	49.47	79.99	81.47	81.85	78.94	80.92	80.29	79.03	79.77
		Kappa ($\kappa \times 100$)									
UH	7x7	88.86	74.33	91.63	88.63	84.87	92.51	91.65	93.25	92.99	95.66
Trento		98.49	82.13	98.55	99.39	98.51	99.30	99.47	99.16	99.23	99.51
MUUFL		85.92	84.03	87.19	87.44	85.54	88.61	89.31	84.44	85.03	91.68
UH	9x9	89.67	66.04	91.90	91.07	83.38	92.76	91.52	93.20	93.53	96.80
Trento		98.47	69.40	98.84	98.90	99.15	99.39	99.50	98.95	98.95	99.57
MUUFL		86.61	80.71	86.41	89.13	88.61	89.50	88.97	84.82	82.73	**92.27**
UH	11x11	90.46	58.25	90.50	90.04	86.73	93.53	92.71	93.81	93.14	**97.07**
Trento		98.40	61.36	98.79	99.26	99.28	98.97	99.40	99.12	98.57	**99.61**
MUUFL		86.97	72.89	86.66	90.97	88.95	90.08	90.46	89.38	81.69	91.13

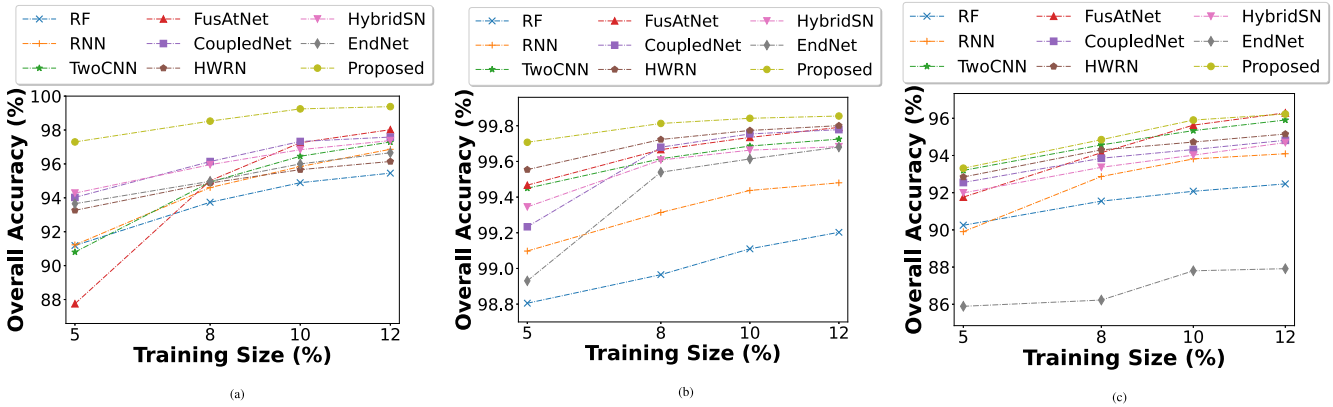


Fig. 10. Overall accuracy (OA) achieved by different methods with varying training sample sizes that are randomly taken from (a) UH, (b) UT, and (c) MUUFL datasets.

samples, i.e., 5%, 8%, 10%, and 12%, respectively, are randomly selected to train the models (both the one proposed in this article and the comparative methods), and the rest of the samples are used for testing purposes. The rest of the parameters remain the same as explained in the previous sections.

The detailed experimental results are presented in Figs. 10–12. From the figures, one can conclude that the experimental results obtained on 5% training samples are inferior to the ones obtained with 12% training samples on all three datasets with all the methods. Figs. 10–12 show that the detailed classification performance using OA, AA, and κ metrics on three datasets with different percentages of randomly selected training samples.

Figs. 10–12 show that the classification accuracies significantly improve as the number of training samples increases, which proves that the spatial–spectral feature learning process proposed in this study is effective for HSI classification while considering the limited availability of training samples. Achieving such higher accuracy and statistical significance while with a limited number of training samples is crucial, especially for HSI classification since acquiring a high number of labeled training samples is often difficult and time-consuming in real scenarios.

F. Ablation Study Along With Statistical Tests

In order to show that the performance of the proposed model is significant and outperformed compared to the state-of-the-art methods, we have used McNemar's statistical significance test [68]. McNemar's test is used to identify whether the experimental results obtained are significantly better or not. Moreover, it is being used to evaluate the statistical significance differences for a sample set, which is calculated based on the standardized normal statistic test

$$Z = \frac{f_{12} - f_{21}}{\sqrt{f_{12} + f_{21}}} \tag{9}$$

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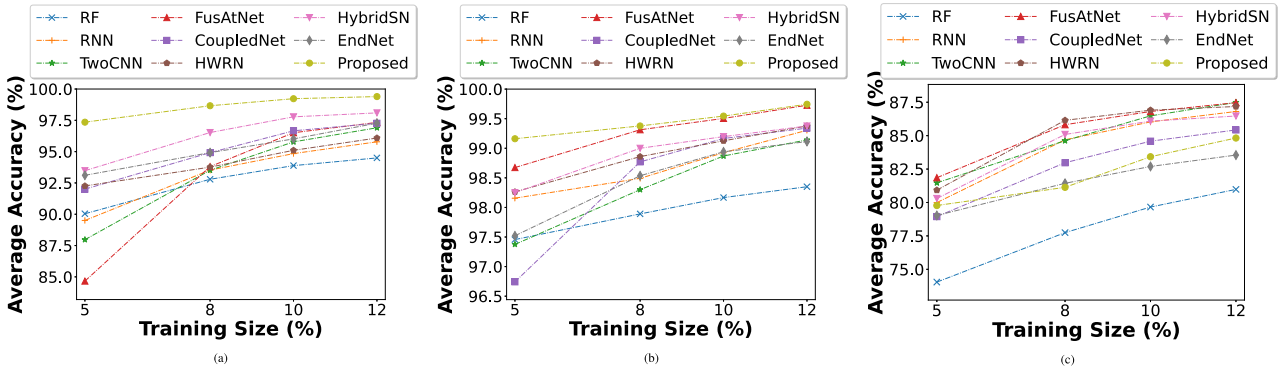


Fig. 11. Average accuracy (AA) achieved by different methods with varying training sample sizes that are randomly taken from (a) UH, (b) UT, and (c) MUUFL datasets.

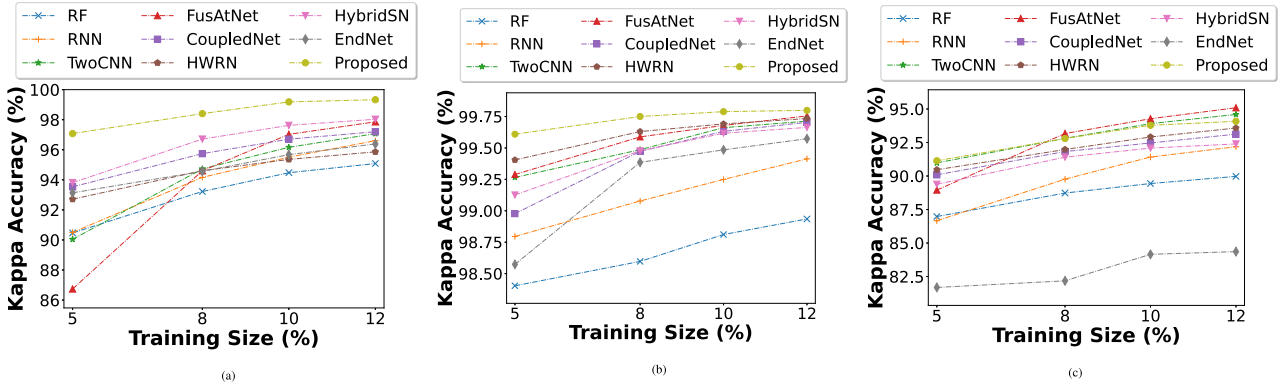


Fig. 12. Kappa (κ) accuracy achieved by different methods with varying training sample sizes that are randomly taken from (a) UH, (b) UT, and (c) MUUFL datasets.

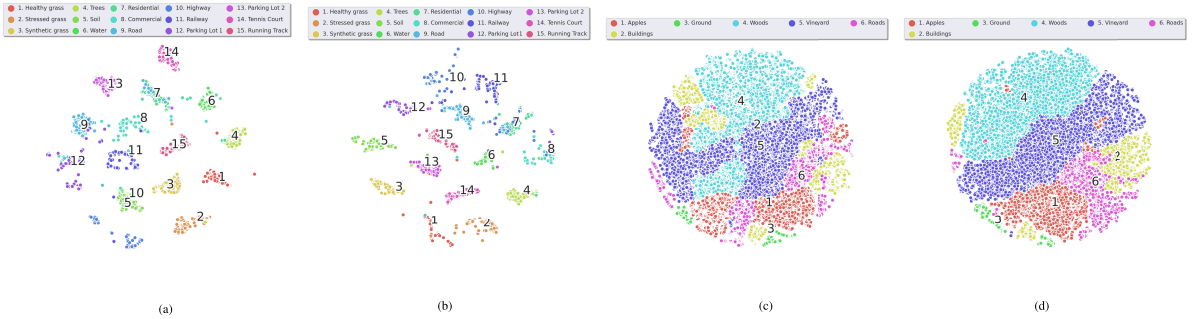


Fig. 13. 2-D feature visualization of the proposed joint spectral–spatial HSIs and LiDAR features via the t-SNE method for disjoint UH and Trento dataset. (a) HSIs. (b) HSIs + LiDAR. (c) HSIs. (d) HSIs + LiDAR.

where f_{12} indicates the number of samples correctly classified by classifier 1 and wrongly classified by the classifier 2. Similarly, f_{21} is the number of samples that can be correctly classified by method 2 but wrongly classified by method 1. The difference between the two methods is said to be statistically significant if $|Z| > 1.96$, whereas $Z > 0$ indicates that method 1 is more discriminative than method 2, and $Z < 0$ means that method 2 is more discriminative than method 1.

Fig. 13 shows the t-distributed stochastic neighbor embedding (t-SNE) process to visualize the 2-D feature spaces learned using the proposed method. t-SNE is usually used to visualize the nonlinear connections among the data samples and, moreover, shows the capacity of any model to learn the nonlinear. t-SNE is relatively helpful when dealing with CNN or any other deep network because deep architectures are usually considered a black box, and there is no way to interpret what is happening at deeper levels of the architecture. Generically, we assume that the deeper level contains the information about more complex objects; however, to some extent, this is not completely true. One can interpret it as explained above; however, HSI datasets are themselves just a high-dimensional noise for individuals [69]. Thus, t-SNE can significantly help us to understand which samples seem similar for the deep model.

Furthermore, the statistical significance values between the proposed method and the compared methods are shown in Table V using the disjoint training samples, i.e., UH and Trento datasets and 5% of training samples for the MUUFL dataset. It is observed that the proposed model performs significantly better for the disjoint datasets compared to the MUUFL dataset, which uses 5% of randomly selected training samples.

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TABLE V

STATISTICAL SIGNIFICANCE OF DIFFERENCE (Z VALUE) BETWEEN DIFFERENT METHODS OVER HSIs AND LiDAR DATASETS

Data (H+L)	Training Ratio	RF [62]	SVM [15]	RNN [63]	Two-CNN [50]	FusAtNet [64]	CoupledNet [8]	HWRN [12]	HybridSN [33]	EndNet [65]
UH	Disjoint	31.56	45.11	37.12	16.52	27.68	31.16	28.57	32.05	24.57
UT		11.84	34.14	37.59	13.03	35.00	16.30	21.91	32.20	20.27
MUUFL	5%	23.97	70.03	26.15	3.47	7.00	9.21	6.38	7.02	47.82

TABLE VI

OA, AA, AND KAPPA VALUES (IN %) ON UH, TRENTO, AND MUUFL DATASETS FOR THE PROPOSED MODEL USING THE VARIOUS HSI AND LiDAR DATA FUSION STRATEGY, I.E., EARLY, MIDDLE, LATE, AND CROSS FUSIONS

Class	UH				Trento				MUUFL			
	Early Fusion	Middle Fusion	Late Fusion	Cross Fusion	Early Fusion	Middle Fusion	Late Fusion	Cross Fusion	Early Fusion	Middle Fusion	Late Fusion	Cross Fusion
1	82.01	83.38	81.76	82.58	97.89	97.30	98.80	98.39	98.13	98.42	98.31	96.49
2	88.28	89.22	82.51	82.76	99.24	99.10	99.30	96.88	88.20	87.88	86.16	85.26
3	97.82	97.62	97.35	92.40	84.04	76.64	62.21	58.73	92.50	90.86	91.23	89.93
4	95.73	95.20	95.64	82.48	99.48	99.46	98.24	98.21	92.62	91.73	90.58	88.85
5	99.87	99.87	99.24	99.08	98.12	96.29	96.65	96.52	93.54	93.59	93.31	92.07
6	79.02	89.51	96.73	82.51	84.89	86.27	83.43	85.13	95.93	97.66	96.53	86.60
7	93.56	92.00	93.84	81.99	/	/	/	/	91.73	90.38	88.41	88.24
8	79.77	89.52	87.05	89.45	/	/	/	/	97.28	96.97	97.46	94.82
9	85.67	86.59	87.69	80.83	/	/	/	/	61.80	57.97	52.68	36.75
10	66.05	64.60	64.57	67.66	/	/	/	/	26.62	25.86	26.43	04.98
11	86.14	82.63	80.10	78.90	/	/	/	/	60.39	63.79	65.35	65.75
12	90.61	90.20	87.15	83.47	/	/	/	/	/	/	/	/
13	82.69	90.52	83.97	79.29	/	/	/	/	/	/	/	/
14	99.59	96.35	99.19	95.27	/	/	/	/	/	/	/	/
15	99.01	99.92	98.87	97.81	/	/	/	/	/	/	/	/
OA	87.82	88.56	87.32	84.07	97.06	96.37	95.86	95.66	94.10	93.81	93.44	91.31
AA	88.39	89.81	89.04	85.10	93.94	92.51	89.77	88.98	81.70	81.37	80.59	75.43
$\kappa(\times 100)$	86.78	87.60	86.29	82.71	96.07	95.15	94.48	94.21	92.19	91.79	91.28	88.50

Moreover, here, we provide an ablation study to further validate the fusion process by using a recently proposed idea [41] related to what, where, and how to fuse, i.e., four different fusion strategies (such as "early," "middle," "late," and "cross" fusion strategies) have been used. More specifically, the fusion process is performed after feature learning using our proposed network. Table VI listed the quantitative comparison between different fusion strategies in terms of OA, AA, and Kappa on HU, Trento, and MUUFL datasets. In general, the feature representations from Hyperspectral and LiDAR data obtained by our proposed feature learning network tend to be fused using the "early" fusion strategy, yielding slightly better classification results on all the experimental datasets compared to those using "middle" and "later" fusion strategies. This possible reason for the phenomenon is that the learned features via the proposed approach are discriminative and "good" enough, and there is no need to use more advanced and complex fusion strategies. The cross-fusion mechanism is well designed in [41] to address the problem of cross-modality learning, which is an end-to-end combination framework, including feature learning and fusion. It should be noted, however, that the proposed feature learning network is more powerful in learning feature representations. This, to some extent, can explain why the fusion results obtained by the cross-fusion strategy are relatively worse than other more simple ones, e.g., "early," "middle," and "late" fusion.

V. CONCLUSION AND FUTURE WORK

A new joint CNN and morphological feature learning framework for HSI and LiDAR data fusion is proposed for accurate land-cover classification. The proposed network learns joint feature representations for HSI and LiDAR data for classification purposes. The proposed model consists of a three-layered CNN and a morphological convolutional network in which the former uses 3-D convolution to effectively extract the spectral–spatial features from HSI data, and the latter can effectively model the complimentary and nonlinear features from LiDAR data by considering morphological dilation and erosion layers, followed by a 2-D convolution with a kernel of size (1×1). To aggregate the most important nonlinear and elevation information, both dilation and erosion layers are learned in parallel. Finally, the spectral–spatial features from HSI data and elevation information from LiDAR data are fused for robust classification. To validate the performance of the proposed network, two datasets, i.e., UH and UT with disjoint training and test samples, and randomly selected training and test samples from MUUFL have been considered for the experimental evaluation. The results show that the proposed model achieves significant performance improvements in terms of OA, AA, and Kappa for both UH, UT, and MUUFL datasets. Future works will focus on developing parallel implementations of the proposed approaches for faster exploitation. In the future, we will also consider the use of more sophisticated morphological filters (such as extended morphological profiles or attribute profiles) instead of the simple dilation and erosion operations for morphological feature extraction.

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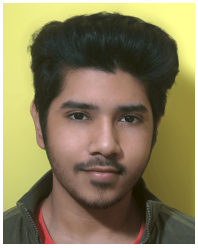
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